

CE725: Vibration-Based Structural Health Monitoring

Homework Project 2

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Question 1: Extended Kalman Filter (EKF)

1. Derivations of the Jacobian Matrices

For joint state-parameter estimation, the standard state vector is augmented with the unknown structural parameters (k_1, k_2, c_1, c_2) . The augmented state vector Z is defined as:

$$Z = [y_1 \ y_2 \ \dot{y}_1 \ \dot{y}_2 \ k_1 \ k_2 \ c_1 \ c_2]^T \quad (1)$$

Continuous Process Equations and Jacobian (A_c)

The continuous-time process model $\dot{Z} = f(Z, \ddot{u}_g)$ defines how the states evolve. Assuming the structural parameters are constant over time ($\dot{k}_1 = 0$, etc.), the non-linear continuous equations are:

$$f_1 = \dot{y}_1 \quad (2)$$

$$f_2 = \dot{y}_2 \quad (3)$$

$$f_3 = \frac{1}{m_1} [-(k_1 + k_2)y_1 + k_2 y_2 - (c_1 + c_2)\dot{y}_1 + c_2 \dot{y}_2] - \ddot{u}_g \quad (4)$$

$$f_4 = \frac{1}{m_2} [k_2 y_1 - k_2 y_2 + c_2 \dot{y}_1 - c_2 \dot{y}_2] - \ddot{u}_g \quad (5)$$

$$f_5 = f_6 = f_7 = f_8 = 0 \quad (6)$$

The continuous process Jacobian matrix is defined as $A_c = \frac{\partial f}{\partial Z}$. Taking the partial derivatives of each function with respect to the 8 states yields an 8×8 matrix with the following non-zero elements:

- $\frac{\partial f_1}{\partial \dot{y}_1} = 1, \quad \frac{\partial f_2}{\partial \dot{y}_2} = 1$

- **Derivatives of f_3 :**

- w.r.t y_1, y_2 : $-(k_1 + k_2)/m_1, \quad k_2/m_1$

- w.r.t $\dot{y}_1, \dot{y}_2$: $-(c_1 + c_2)/m_1, \quad c_2/m_1$

- w.r.t k_1, k_2, c_1, c_2 : $-y_1/m_1, \quad -(y_1 - y_2)/m_1, \quad -\dot{y}_1/m_1, \quad -(\dot{y}_1 - \dot{y}_2)/m_1$

- **Derivatives of f_4 :**

- w.r.t y_1, y_2 : $k_2/m_2, \quad -k_2/m_2$

- w.r.t $\dot{y}_1, \dot{y}_2$: $c_2/m_2, \quad -c_2/m_2$

– w.r.t k_2, c_2 : $(y_1 - y_2)/m_2, \quad (\dot{y}_1 - \dot{y}_2)/m_2$

Note: The discrete state-transition Jacobian F_k used in the EKF covariance prediction is obtained by applying the chain rule through the two RK2 integration steps (Ralston's method) utilizing A_c .

Measurement Equation and Jacobian (G_k)

The accelerometers measure the absolute acceleration of the stories $(\ddot{y} + \ddot{u}_g)$. The non-linear measurement function $H(Z)$ is extracted from f_3 and f_4 by shifting the ground motion term:

$$h_1(Z) = \frac{1}{m_1} [-(k_1 + k_2)y_1 + k_2y_2 - (c_1 + c_2)\dot{y}_1 + c_2\dot{y}_2] \quad (7)$$

$$h_2(Z) = \frac{1}{m_2} [k_2y_1 - k_2y_2 + c_2\dot{y}_1 - c_2\dot{y}_2] \quad (8)$$

The measurement Jacobian $G_k = \frac{\partial H}{\partial Z}$ is a 2×8 matrix. Because $H(Z)$ consists of the acceleration components of $f(Z)$, G_k is identical to rows 3 and 4 of the A_c matrix derived above.

2. Effect of Initial Covariance (P_0) and Process Noise (Q)

The performance of the EKF is highly sensitive to the tuning of its noise matrices:

- **Initial Covariance Matrix ($P_{0,0}$):** This matrix signifies the confidence in the initial state guess ($Z_{0,0}$). High values indicate high uncertainty, causing the filter to heavily weight incoming measurements for rapid correction; however, this can lead to erratic initial oscillations. Conversely, low values instruct the filter to trust the initial guess, resulting in a sluggish response that takes longer to correct if the initial guess is inaccurate.
- **Process Noise Covariance Matrix (Q):** This matrix accounts for unmodeled dynamics and mathematical inaccuracies. High values assume an untrustworthy process model, relying more on noisy measurements to quickly track sudden changes (at the cost of jittery estimates). Low values assume high model accuracy, yielding smooth estimates but limiting the filter's ability to track actual physical changes in the structure.

3. Importance of Process Noise for Joint Estimation

In this problem, the structure experiences a sudden stiffness drop at story 1 at $t = 10$ seconds. In the augmented state equations, the parameters are modeled as strict constants ($\dot{k}_1 = 0, \dot{c}_1 = 0$, etc.).

If the process noise corresponding to these parameters is set to zero ($Q_{5,5} = 0$), the EKF will continually reduce the covariance for these states over time. Eventually, the filter becomes completely "certain" of its parameter estimates, causing the corresponding Kalman Gain to approach zero. Consequently, when the damage occurs at 10 seconds, the filter will attribute the new measurement discrepancy entirely to measurement noise or state variations, failing to recognize that the parameter itself has changed.

By injecting process noise ($Q > 0$) for the parameter states, the covariance is artificially prevented from reaching zero. This ensures the Kalman Gain remains active, keeping the filter responsive and allowing it to detect and track the sudden change in stiffness.

4. Analysis and Observations

Base State and Parameter Estimation

The EKF successfully filters the noisy acceleration signals to reconstruct the unobserved displacement and velocity states with high fidelity. For the structural parameters, the filter correctly identifies the discrete stiffness degradation in k_1 from 10,000 N/m to 7,500 N/m at $t=10$ seconds. A brief convergence delay is observed immediately following the damage event; this is expected behavior, as the algorithm requires a few time steps to process the new measurement innovation sequence and adjust the state estimates accordingly.

Sensitivity to Initial Covariance (P_0)

By evaluating the four choices of P_0 from Table 2, distinct transient responses were observed during the initial seconds of the simulation. Choice 1 (high parameter variance, 1010) resulted in severe initial spikes, as the filter completely distrusted the initial guess and overcompensated using early noisy measurements. Conversely, Choice 4 (extremely low variance) produced a sluggish, overly damped response, largely ignoring initial sensor data. Choices 2 and 3 provided an optimal balance, converging smoothly without excessive early volatility.

Sensitivity to Process Noise (Q)

The simulation across the three Q matrix choices from Table 3 clearly demonstrates the filter's dependence on process noise for structural health monitoring. When applying Choice 1 ($Q=0$), the filter completely failed to detect the damage at $t=10$ seconds, maintaining a flat estimate of 10,000 N/m. Choice 3 (high process noise) allowed the filter to track the sudden stiffness drop almost instantaneously, but at the cost of highly volatile, jagged parameter estimates throughout the time history. Choice 2 offered the most realistic performance, effectively tracking the damage with a smooth, albeit slightly slower, convergence.

Question 2: Unscented Kalman Filter (UKF)

1. UKF Implementation Process

While the Extended Kalman Filter (EKF) relies on the first-order analytical linearization of the system dynamics (via Jacobian matrices), the Unscented Kalman Filter (UKF) employs the Unscented Transform to handle system non-linearities. The process implemented for this solution is as follows:

- **Sigma Point Generation:** For the augmented state vector of dimension $N = 8$, a deterministic set of $2N + 1 = 17$ sigma points was generated at each time step. These points were calculated using the Cholesky factorization of the current error covariance matrix to ensure they precisely capture the true mean and covariance of the states.
- **Parameter Scaling and Weights:** The spread and weighting of the sigma points were governed by the prescribed scaling parameters: $\lambda = -7.97$, $\alpha = 0.1$, and $\beta = 0$. The specific weights applied for the mean ($W^{(m)}$) and covariance ($W^{(c)}$) were:

$$\begin{aligned} W_0^{(m)} &= -265.67 \\ W_0^{(c)} &= -262.68 \\ W_i^{(m)} &= W_i^{(c)} = 16.667 \quad \forall i = 1, \dots, 16 \end{aligned}$$

- **Non-linear Propagation:** Instead of linearizing the process and measurement models, each of the 17 sigma points was propagated directly through the true non-linear continuous-time equations using Ralston's integration method. The predicted state mean, predicted measurement, and their respective covariances were then reconstructed by taking the weighted sum of these transformed points.

2. Observations and Filter Comparison

Based on the overlaid parameter estimation plots, the following comparisons and observations can be made between the EKF and UKF approaches:

- **Analytical vs. Deterministic Handling:** The most significant operational difference is that the UKF completely bypassed the need to derive and compute the continuous process Jacobian (A_c) and measurement Jacobian (G_k). By treating the non-linear equations as a "black box" for the sigma points, the UKF eliminates the risk of analytical derivation errors, which is highly advantageous for complex structural systems.
- **Tracking Performance (Accuracy):** Theoretically, the UKF is superior as it captures the posterior mean and covariance accurately to the 2nd order of the Taylor series expansion, whereas the EKF only captures it to the 1st order. However, observing the tracking of the k_1 stiffness degradation at $t = 10$ seconds, both the UKF and EKF exhibit nearly identical trajectories.
- **Conclusion:** The high degree of overlap between the EKF and UKF estimates indicates that the non-linearities present in this specific 2-DOF shear building model are relatively mild during propagation. Consequently, the first-order approximation made by the EKF is sufficient to yield performance on par with the second-order accuracy of the UKF for this specific structural health monitoring application.